


1.

Perpetuity: A constant stream of cash flows that continues forever

Growing Perpetuity: A stream of cash flows that grows at a constant rate forever

Annuity: A series of equal cash flows paid over a fixed number of periods

Growing annuity: A series of cash flows that grows at a constant rate over a fixed number of periods

2.

The three common forms of business organizations are:

1. Sole Proprietorship

2. Partnership

3. Corporation

Three key features of a corporation are:

1. Limited liability

2. Transferability of ownership

3. Separate legal entity

3.

When the interest rate r increase:

The future value of an annuity increases because of higher compounding returns

The present value decreases because future cash flows are discounted at a higher rate

4.

I disagree with the statement.

The current stock price reflects the present value of expected future cash flows, including long-term profits.

Therefore, maximizing stock value does not necessarily mean maximizing short-term profits

However, manager should avoid short-term actions that damage the firm's long-term value.

Part II

1.

$$\begin{aligned} \text{EBIT} &= 19800 - 10900 - 2100 \\ &= 6800 \end{aligned}$$

$$\begin{aligned} \text{Taxes} &= (6800 - 1250) \times 40\% \\ &= 2220 \end{aligned}$$

$$\begin{aligned} \text{OCF} &= \text{EBIT} + \text{Depreciation} - \text{Taxes} \\ &= 6800 + 2100 - 2220 \\ &= 6680 \end{aligned}$$

2.

(1)

$$\begin{aligned} \text{FV} &= 11827, r = 7\%, T = 8 \\ \text{PV} &= \frac{11827}{(1.07)^8} \\ &= 8047.44 \end{aligned}$$

(2)

$$\begin{aligned} \text{FV} &= 725380, r = 11\%, T = 17 \\ \text{PV} &= \frac{725380}{(1.11)^{17}} \\ &= 123048.11 \end{aligned}$$

$$\begin{aligned} (3) \quad \text{PV} &= 242, \text{FV} = 345, T = 4 \\ r &= \left(\frac{345}{242} \right)^{\frac{1}{4}} - 1 \\ &= 9.21\% \end{aligned}$$

$$\begin{aligned} (4) \quad \text{PV} &= 410, \text{FV} = 927, T = 8 \\ r &= \left(\frac{927}{410} \right)^{\frac{1}{8}} - 1 \\ &= 10.74\% \end{aligned}$$

(5)

$$PV = 625, FV = 1254, k = 7\%$$

$$T = \frac{\ln \left(\frac{1254}{625} \right)}{\ln(1.07)}$$

$$\approx 0.64 \text{ Years}$$

(6)

$$PV = 810, FV = 4341, k = 12\%$$

$$T = \frac{\ln \left(\frac{4341}{810} \right)}{\ln(1.12)}$$

$$\approx 14.81 \text{ Years}$$

3.

(a)

Set NPV equal to zero

$$0 = 9400 - \frac{4500}{1+IRR} - \frac{3100}{(1+IRR)^2} - \frac{2400}{(1+IRR)^3} - \frac{1800}{(1+IRR)^4}$$

$$IRR = 11.65\%$$

(b)

$$10\% < IRR = 11.65\%$$

Reject the offer

(c)

$$20\% > IRR = 11.65\%$$

Accept the offer

(d)

$$NPV_{10\%} = 9400 - \frac{4500}{1.1} - \frac{3100}{1.1^2} - \frac{2400}{1.1^3} - \frac{1800}{1.1^4} = -285.47$$

$$NPV_{20\%} = 9400 - \frac{4500}{1.2} - \frac{3100}{1.2^2} - \frac{2400}{1.2^3} - \frac{1800}{1.2^4} = 1240.28$$

(e)

Yes. The IRR rule is reversed because this is a financing-type cash flow.

4.

$$D = \frac{1650000}{3} = 550000$$

$$\begin{aligned} EBIT &= 1240000 - 485000 - 550000 \\ &= 205000 \end{aligned}$$

$$\begin{aligned} OCF &= EBIT(1-T) + D \\ &= 205000(0.65) + 550000 \\ &= 683250 \end{aligned}$$

$$NPV = -1650000 + 683250 \left[\frac{1 - (1.12)^3}{0.12} \right] = -89488.79$$

Decision: Reject the Project because NPV < 0.

5.

$$a) h = \frac{585000}{8} = 73125$$

$$\begin{aligned} Q_{BE} &= \frac{FC + h}{p - v} \\ &= \frac{695000 + 73500}{31 - 20} \\ &= 48032 \text{ units} \end{aligned}$$

at least 48032 units must be sold to break even

b)

$$\text{Sales} = 20000 \times 36 = 720000$$

$$VC = 20000 \times 20 = 400000$$

$$\begin{aligned} EBIT &= 720000 - 400000 - 695000 - 73500 \\ &= -351500 \end{aligned}$$

$$\begin{aligned} OCF &= EBIT(1-T) + D \\ &= -351500(0.65) + 73500 \\ &= -191975 \end{aligned}$$

$$NPV = -588000 + 39915 \left[\frac{1 - (0.15)^8}{0.15} \right] = 767058.91$$

$$\begin{aligned} \frac{\partial NPV}{\partial a} &= (P - V)(1 - T) \times PVIFA \\ &= 1660.65 \left[\frac{1 - (0.15)^8}{0.15} \right] \\ &= 46.67 \text{ per unit} \end{aligned}$$

$$\begin{aligned} \Delta NPV &= -500 (46.67) \\ &= -23335 \end{aligned}$$

$$NPV_{\text{new}} = 743723.91$$

The Project still has a positive NPV

(c)

$$OCF = [Q(P - V) - FC - D](1 - T) + bD$$

$$\begin{aligned} \frac{\partial OCF}{\partial V} &= -Q(1 - T) \\ &= -70000(0.35) \\ &= -24500 \end{aligned}$$